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Machine alerts need understandable context.

The technician’s question

What is happening with this pump, which signals matter, and what can I trust in the returned analysis?

The product boundary

A spoken view of local telemetry and a separate request for model review. The operator remains responsible for the next decision.

For plant operators and maintenance technicians. A future plant pilot would measure investigation time and comprehension.

Physical signals.
Two explicit request paths.

Check pump P-204

The ESP32-C3 emulates NORMAL, WARNING and CRITICAL. Local rules assess fresh values. A status read makes no AMD inference request.

Analyze pump P-204

The existing Xiaozhi / AssemblyAI bridge calls the industrial Core contract. X6 routes explicit analysis to Qwen3-32B on one AMD MI300X.

All pump signals are explicitly simulated. The governed cloud route and returned model executions are real.

A result is useful when its scope stays honest.

Bounded model review

Local policy establishes severity. Model output must satisfy a strict contract and the human-inspection allowlist. No physical actuation is authorized.

Historical means historical

Newer telemetry supersedes the admitted sample. Recorded AMD responses are historical, and current recommendations are suppressed.

Interruption suppresses obsolete replies. Reconnection does not replay execution. Firmware neither holds cloud credentials nor selects a GPU.

Recorded execution.
Inspectable boundaries.

WARNING: 6.128 seconds

A separate recorded backend run passed assessment on Qwen3-32B / one MI300X. Its result scope is historical_sample.

CRITICAL: 8.466 seconds

A second recorded backend run passed assessment on the same model and deployment. Its result scope is historical_sample.

These are individual observations, not benchmark guarantees. Hardware identity is deployment inventory. Acoustic return has an operator listening declaration, not a firmware ACK.

Build on the existing Core.
Keep decisions with people.

FORGE contribution

Industrial contracts, the physical C3 emulator, telemetry ingest, a recommendation gate and evidence capture. Core/X6 and voice transport predate FORGE.

Next: a measured pilot

Measure operator comprehension and investigation time in a defined maintenance workflow. Broader automation needs its own authorization and execution contracts.

jarvisprometheus.space/forge/ - recorded prototype, source and evidence. Autonomous GPU selection, physical failover and eight-GPU execution are not demonstrated.